Insight

Agent 001

The Old Is New Again

We take an early peek at the new graphics chipset from NVIDIA. The GTX 200 series is big and fast, we find out what makes it tick

There are war clouds on the horizon. To our regular readers the ongoing battle of words and rattling of silicon sabers between NVIDIA and Intel would not come as news. Intel is set to unleash its Larrabee processors in the near future which will go against the graphic accelerators of today; as manufactured by NVIDIA and AMD/ATI. Meanwhile, the other side has been adding General Purpose to their GPUs for a while now; the so-called GP-GPUs which even today are all set to take over tasks traditionally executed by the CPU. On a more general level, the entire industry is moving towards massive parallelization—instead of discrete processors performing specific tasks, we are looking at processors that can wear multiple hats and churn everything from physics, to sound, to graphics.

But let's limit the battleground to the CPU vs. GPU scene, and as a subset to the diametrically opposite directions NVIDIA and ATI are taking with their new generation hardware.

Meet The New Boss

The G80/G92 processors from NVIDIA have been unchallenged both in pure performance and in the price-to-performance ratio the past years. There has simply not been a better card to get than one based on either the 8800-series or the 9800-series. The G80 enjoyed a nice stint on the throne.

NVIDIA would like to repeat that success with their just-released GTX 200 hardware. In many ways the new graphics chip from the green team is a consolidation of their past successes and their future aspirations. The GTX 200 is essentially a more efficient G80/G92 core, and is simultaneously a bigger version of its successful sibling. At 1.4 billion transistors, the new core is a monster, and using a 65nm fabrication process, NVIDIA claims that the GTX 200 is the largest and the most complicated piece of silicon that they have yet produced.

NVIDIA's ambitions with the GTX 200 chips is just as large: not only do they plan to retain their crown in the graphics industry, but they also hope to make substantial inroads into the traditional CPU market with this new chip. Somewhat amusingly, NVIDIA terms these twin goals as Gaming Beyond and Beyond Gaming. Marketing monikers aside, let's take a close look at the changes wrought through this new chip by dissecting its internals.

(Almost) Same As The Old Boss

The heart of the GTX 200 beats with the same chewy good-

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